

Summary of Trigonometric Identities & Calculus Concepts

Pythagorean Identity

The fundamental trigonometric identity states:

$$\sin^2 y + \cos^2 y = 1$$

This identity holds for all values of y and can be derived from the Pythagorean theorem applied to a unit circle.

Finding Secant, Cosecant, and Cotangent

These are the reciprocal trigonometric functions:

- Secant (sec):

$$\sec y = \frac{1}{\cos y}$$

- Cosecant (csc):

$$\csc y = \frac{1}{\sin y}$$

- Cotangent (cot):

$$\cot y = \frac{1}{\tan y} = \frac{\cos y}{\sin y}$$

Derivatives of Trigonometric Functions

- $\frac{d}{dx} \sin x = \cos x$
- $\frac{d}{dx} \cos x = -\sin x$
- $\frac{d}{dx} \tan x = \sec^2 x$
- $\frac{d}{dx} \cot x = -\csc^2 x$
- $\frac{d}{dx} \sec x = \sec x \tan x$
- $\frac{d}{dx} \csc x = -\csc x \cot x$

Integrals of Trigonometric Functions

- $\int \sin x \, dx = -\cos x + C$
- $\int \cos x \, dx = \sin x + C$

- $\int \sec^2 x \, dx = \tan x + C$
- $\int \csc^2 x \, dx = -\cot x + C$
- $\int \sec x \tan x \, dx = \sec x + C$
- $\int \csc x \cot x \, dx = -\csc x + C$

Methods for Solving Integration & Differentiation

1. **Basic Rules** – Power rule, product rule, quotient rule, and chain rule.
2. **Substitution Method** – Used for integrals where substitution simplifies the function.
3. **Integration by Parts** – Based on the formula:

$$\int u \, dv = uv - \int v \, du$$

4. **Trigonometric Identities** – Useful in simplifying integrals involving trigonometric functions.
5. **Partial Fractions** – Used when the integral involves rational expressions.
6. **Trigonometric Substitution** – Used for integrals involving expressions like $\sqrt{a^2 - x^2}$.
7. **Differentiation via Implicit Differentiation** – Used for functions where y is not explicitly solved for x .
8. **Numerical Integration** – Methods like Trapezoidal Rule and Simpson's Rule for approximations.

Let me know if you need a deeper explanation of any of these! 🚀



explain more like add the chain rule, substitution and many other ways

Comprehensive Guide to Trigonometric Identities, Differentiation, and Integration Methods

1. Fundamental Trigonometric Identity

One of the most important identities in trigonometry is:

$$\sin^2 y + \cos^2 y = 1$$

This identity is derived from the Pythagorean theorem applied to the unit circle. Using this identity, we can derive two other useful forms:

$$1 + \tan^2 y = \sec^2 y$$

$$1 + \cot^2 y = \csc^2 y$$

These are used in calculus to simplify expressions involving tangent and cotangent.

2. Reciprocal Trigonometric Functions

To find secant, cosecant, and cotangent, we use their reciprocal definitions:

- Secant: $\sec y = \frac{1}{\cos y}$
- Cosecant: $\csc y = \frac{1}{\sin y}$
- Cotangent: $\cot y = \frac{\cos y}{\sin y}$

3. Differentiation of Trigonometric Functions

Here are the derivatives of basic trigonometric functions:

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \cot x = -\csc^2 x$$

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\frac{d}{dx} \csc x = -\csc x \cot x$$

4. Rules and Techniques for Differentiation

a) Chain Rule

If a function is composed of another function, use:

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

Example:

$$\frac{d}{dx} \sin(3x) = \cos(3x) \cdot 3 = 3 \cos(3x)$$

b) Product Rule

If two functions are multiplied, use:

$$\frac{d}{dx} [f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

Example:

$$\frac{d}{dx} [x \sin x] = 1 \cdot \sin x + x \cos x$$

c) Quotient Rule

If one function is divided by another:

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$$

Example:

$$\frac{d}{dx} \left(\frac{\sin x}{x} \right) = \frac{\cos x \cdot x - \sin x \cdot 1}{x^2}$$

d) Implicit Differentiation

Used when y is a function of x , but not explicitly solved for x .

Example:

If $x^2 + y^2 = 1$, differentiate both sides:

$$2x + 2y \frac{dy}{dx} = 0$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = -\frac{x}{y}$$

5. Integration of Trigonometric Functions

Here are the basic trigonometric integrals:

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

6. Techniques for Solving Integrals

a) Substitution Method

Used when an integral contains a function and its derivative.

Example:

$$\int \sin(2x) \cos(2x) dx$$

Let $u = \sin(2x)$, then $du = 2 \cos(2x)dx$.

Rewriting the integral:

$$\frac{1}{2} \int u du = \frac{1}{2} \frac{u^2}{2} = \frac{\sin^2(2x)}{4} + C$$

b) Integration by Parts

Uses the formula:

$$\int u dv = uv - \int v du$$

Example:

$$\int x \cos x dx$$

Let $u = x$, so $du = dx$.

Let $dv = \cos x dx$, so $v = \sin x$.

Applying the formula:

$$x \sin x - \int \sin x dx = x \sin x + \cos x + C$$

c) Trigonometric Substitution

Used for integrals involving square roots.

For $\sqrt{a^2 - x^2}$, substitute $x = a \sin \theta$.

Example:

$$\int \frac{dx}{\sqrt{4 - x^2}}$$

Let $x = 2 \sin \theta$, then $dx = 2 \cos \theta d\theta$.

Substituting:

$$\begin{aligned} & \int \frac{2 \cos \theta d\theta}{\sqrt{4 - 4 \sin^2 \theta}} \\ &= \int \frac{2 \cos \theta d\theta}{2 \cos \theta} = \int d\theta = \theta + C \end{aligned}$$

Since $\theta = \sin^{-1}(x/2)$:

$$\int \frac{dx}{\sqrt{4 - x^2}} = \sin^{-1}\left(\frac{x}{2}\right) + C$$

d) Partial Fractions

Used when integrating rational functions.

Example:

$$\int \frac{1}{x^2 - 1} dx$$

Factor $x^2 - 1 = (x - 1)(x + 1)$, and write:

$$\frac{1}{x^2 - 1} = \frac{A}{x - 1} + \frac{B}{x + 1}$$

Solving for A and B , then integrating separately.

7. Advanced Methods

- **Taylor Series Expansion** – Used to approximate integrals.
- **Fourier Series** – Used for periodic functions.
- **Laplace Transforms** – Converts a function into an algebraic equation.

Conclusion

By mastering these techniques, you can solve complex differentiation and integration problems with trigonometric functions. Let me know if you need deeper explanations on any topic! 🚀

whats arcsin or arctan

Arcsin and **Arctan** are the inverse functions of **sin** and **tan**, respectively. They are used to find the angle whose sine or tangent is a given value.

1. Arcsin (or $\sin^{-1}$)

- The **arcsin** function gives the angle whose sine is a specific value.
- Denoted as $\arcsin(x)$ or $\sin^{-1}(x)$.
- If $y = \arcsin(x)$, then $\sin(y) = x$, and y will be within the range $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ (i.e., between -90° and 90°).

Example:

$$y = \arcsin(0.5) \quad \text{means} \quad \sin(y) = 0.5$$

The angle y is 30° (or $\frac{\pi}{6}$ radians), because $\sin(30^\circ) = 0.5$.

2. Arctan (or $\tan^{-1}$)

- The **arctan** function gives the angle whose tangent is a specific value.
- Denoted as $\arctan(x)$ or $\tan^{-1}(x)$.
- If $y = \arctan(x)$, then $\tan(y) = x$, and y will be within the range $-\frac{\pi}{2} < y < \frac{\pi}{2}$ (i.e., between -90° and 90°).

Example:

$$y = \arctan(1) \quad \text{means} \quad \tan(y) = 1$$

The angle y is 45° (or $\frac{\pi}{4}$ radians), because $\tan(45^\circ) = 1$.

General Notes:

- The inverse trigonometric functions are important for finding angles when you know the ratio of sides of a right triangle.
- The outputs (angles) of **arcsin** and **arctan** are always restricted to specific ranges to ensure the function is one-to-one (i.e., each input corresponds to exactly one output).

Let me know if you'd like a deeper dive into their applications